MAKESHIFT BASE CHANGE: FOR A FLAT MORPHISM THE TWO SHRIEKS AGREE ON AFFINE OPENS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file duality.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with an input of the shared preamble and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-duality/preamble.tex is excerpted verbatim under the recipe fixed by LANE_BRIEF 22.1, as three spans, lines 8-13 (verbatim package and the reference/slogan/history environments), lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. The counted statement and proof contain no cross reference, no citation and no included file, so the delivered body is self contained.

Editorial note. Canonical location of the counted proposition: Stacks tag 0BZY (sources/stacks-duality/pinned-original-source.tex line 5094).

Allowed standard prerequisites (editorial).

- The local calculation of the shriek on affines. Same chapter, Remark labelled remark-local-calculation-shriek, cited at proof lines 5137-5142. Identifies the two sheaves with the algebraic objects.
- The algebraic base change for the relative dualizing complex in the flat case. Dualizing Complexes, Section labelled dualizing-section-relative-dualizing-complex-algebraic and Lemma labelled dualizing-lemma-bc-flat, cited at proof lines 5142-5150. Supplies the isomorphism after the affine translation.

Source reference. Original reference 0F42: sources/stacks-duality/pinned-original-source.tex, source lines 3953–3953. Same-chapter pointer into the author environment situation at source line 3953; invoked at proof line None of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original reference 0BV2: sources/stacks-duality/pinned-original-source.tex, source lines 4636–4636. Same-chapter pointer into the author environment remark at source line 4636; invoked at proof line 5118 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original section 0BZI: sources/stacks-chow/tags.txt, source lines 15051–15051. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 5124 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Date: stacks-project:duality.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 0BZS: sources/stacks-chow/tags.txt, source lines 15062–15062. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 5126 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Lemma 0.1 (Makeshift base change)**.** *In Situation 0F42 let*

$$\begin{array}{ccc} X' & \xrightarrow{g'} & X \\ {\scriptstyle f'}\downarrow & & \downarrow{\scriptstyle f} \\ Y' & \xrightarrow{g} & Y \end{array}$$

be a cartesian diagram of FTS_S. Let $E \in D^+_{QCoh}(\mathcal{O}_Y)$ be an object such that Lg^*E is in $D^+(\mathcal{O}_Y)$. If f is flat, then $L(g')^*f^!E$ and $(f')^!Lg^*E$ restrict to isomorphic objects of $D(\mathcal{O}_{U'})$ for $U' \subset X'$ affine open mapping into affine opens of Y, Y', and X.

Proof. By our assumptions we immediately reduce to the case where X, Y, Y', and X' are affine. Say $Y = \mathrm{Spec}(R)$, $Y' = \mathrm{Spec}(R')$, $X = \mathrm{Spec}(A)$, and $X' = \mathrm{Spec}(A')$. Then $A' = A \otimes_R R'$. Let E correspond to $K \in D^+(R)$. Denoting $\varphi : R \to A$ and $\varphi' : R' \to A'$ the given maps we see from Remark 0BV2 that $L(g')^*f^!E$ and $(f')^!Lg^*E$ correspond to $\varphi^!(K) \otimes^{\mathbf{L}}_A A'$ and $(\varphi')^!(K \otimes^{\mathbf{L}}_R R')$ where $\varphi^!$ and $(\varphi')^!$ are the functors from Dualizing Complexes, Section 0BZI. The result follows from Dualizing Complexes, Lemma 0BZS. □